

Evaluating Machine Learning Potentials of the Open Catalyst Project

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Introduction

Catalysts speed up reactions that turn harmful gases into useful chemicals, crucial for green chemistry and climate solutions. Traditional methods for studying reactions are slow and expensive. The **Open Catalyst Project (OCP)** uses **ML** to rapidly predict catalyst-molecule interactions with a relatively much lower cost. This study assesses **OCP's UMA-S-1 model** on the absorption of PEG on a Zinc surface to observe and advance catalyst design.

Research Question

How accurately can the **OCP's UMA-S-1 model** predict the absorption of **polyethylene glycol (PEG)** on a **Zinc (Zn)** surface?

Methods & Materials

We used **Python** to run the **UMA-S-1** model via **FAIRChem**, accessing OCP's ML data, scripts, and model architectures within FAIRChem's environment.

Results & Discussion

- The model largely appears **fine**.
- The bond angles are **v-shaped**.
- The values produced seem **reasonable** and **realistic**.
- The bond lengths **seem reasonable as well**.
- The Polyethylene Glycol (PEG) molecules **drift away** from the Zinc Slab.
- **Non-ML** optimization with Density Functional Theory (DFT) takes **24 hours** with **48 cpu cores** while **UMA-S-1** with molecular dynamics (MD) takes less than **10 minutes** with approximately **8 cpu cores**.

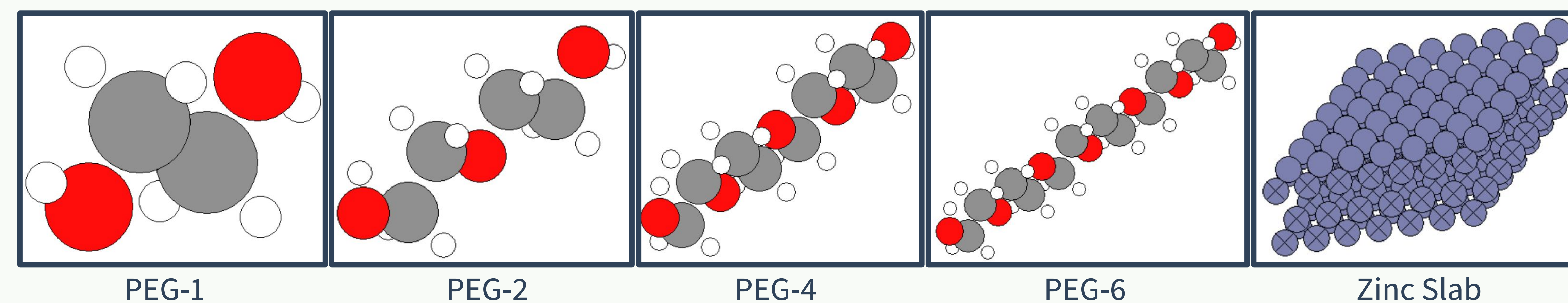
Conclusions

- While UMA-S-1 demonstrates **reliable performance** on **OC20-like systems**, particularly for small adsorbates such as H, CO, and HCOOH on transition metal surfaces, its **limitations** become apparent when applied to molecules **beyond its frequent training domain**.
- The model may output energies **relative to an internal baseline**, requiring **correction** before comparison with DFT.
- For **larger, flexible molecules** like PEG or HCOO*, especially on surfaces like Zn(111), UMA-S-1's **cost noticeably increases**.
- Results from runs testing **PEG on various zinc surfaces** display the model's **strong performance** in terms of **calculation speed and cpu usage**, but its inaccuracies highlight the importance of **further caution and training** when utilized for predictions of PEG absorptions on the Zinc slab.

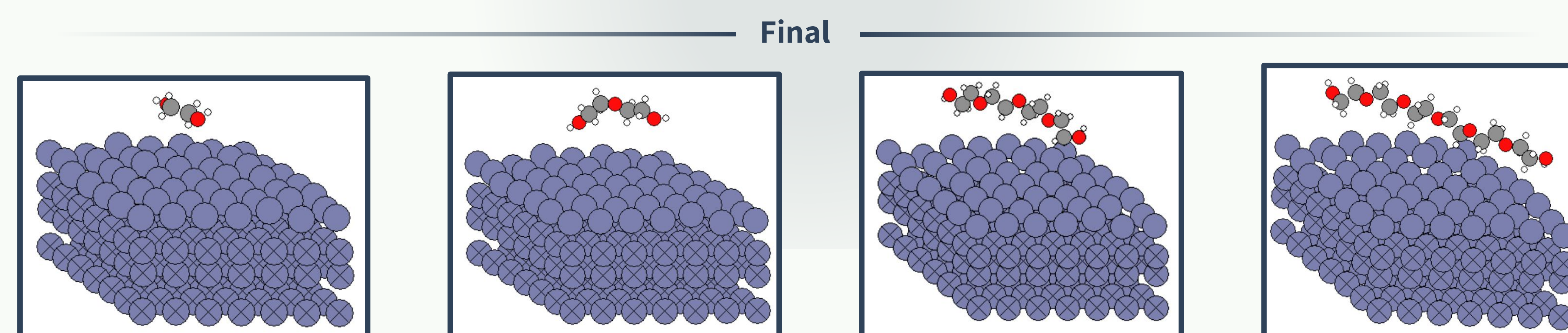
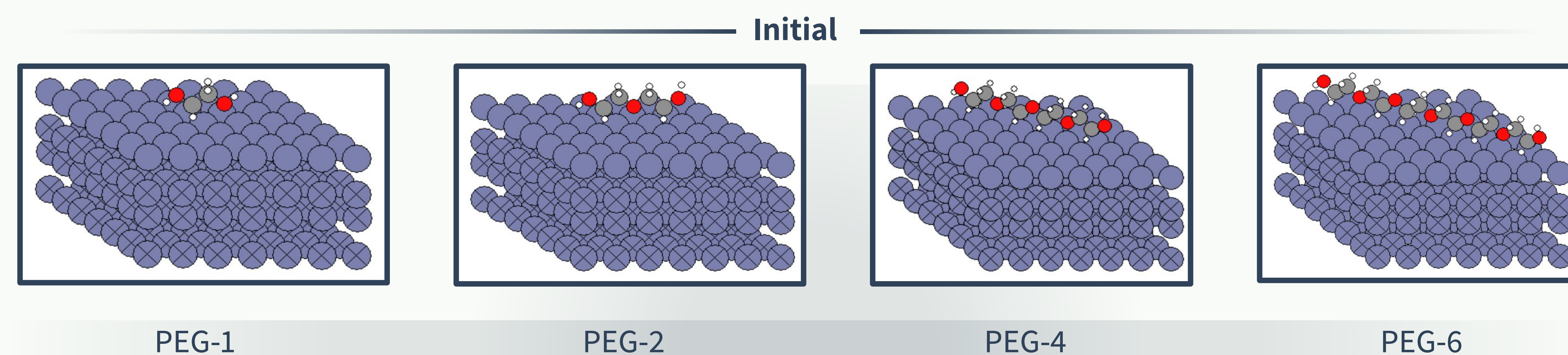
Acknowledgements

We'd like to thank **Dr. Wenrui Chai** for his guidance, instruction, and support through this project, and the **Fundamental AI Research at Meta and Carnegie Mellon University** for providing the datasets and pre-trained models that made this research possible.

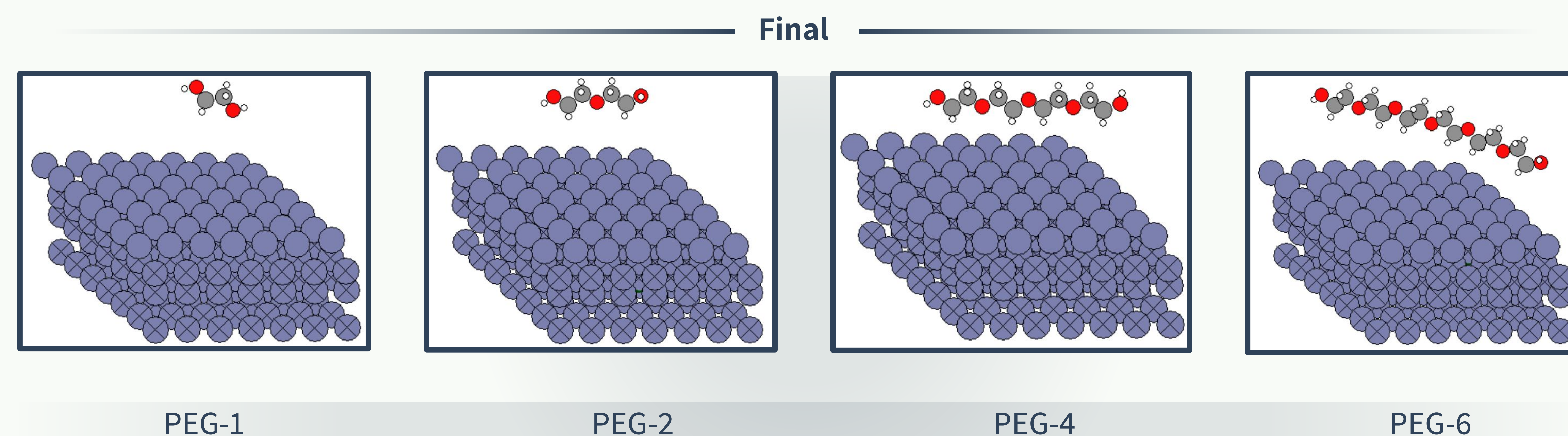
Optimized PEGs and Zinc Slab



MD on Zinc Slab



L'BFGS Line Search on Zinc Slab (Same Initial Positions)



References

S. S. Schoenholz, E. D. Cubuk, et al. (2020). The Open Catalyst 2020 (OC20) Dataset and Community Challenges. ACS Catalysis, 11(10), 6059–6072. <https://doi.org/10.1021/acscatal.0c04525>